

Sample PDF

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This PDF is three pages long. Three long pages. Or three short pages if you're optimistic. Is it the same as saying "three long minutes", knowing that all minutes are the same duration, and one cannot possibly be longer than the other? If these pages are all the same size, can one possibly be longer than the other?

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Python programming

Data Structure

There are four types of data structure as given below:

1. List
2. Tuple
3. Sets
4. Dictionaries

➤ Now explain these types

1. List :-

List is a data structure to store multiple values in a single variable. List is mutable which means that we can modify it after making list. We can store different data type in a list enclosed in square brackets [] and separated by commas.

Example:

```
students = ["Ahmed", "Abdul Rehman", "Rayan"]  
print(students)
```

 Output: ["Ahmed", "Abdul Rehman", "Rayan"]

We can also access ,modify, and perform different operations

◆ Accessing List Items:



Use indexes (starting from 0):

```
fruits = ["Mango", "Apple", "Banana"]  
print(fruits[1])    #output :Apple
```

◆ Modifying List:

```
fruits[0] = "Orange"  
fruits.append("Pineapple")  
print(fruits)
```

Output: ['Orange', 'Apple', 'Banana', 'Pineapple']

◆ Common List Methods:

Method	Function
append()	Adds an item to the end
remove()	Removes an item
sort()	Sorts the list
reverse()	Reverses the list order

➤ More operations in list (slicing + concatenation)

Slicing:

```
List2 = [1,2,3,4,5,6,7,8,9,10,11]
```

```
(list2[1:11:2])    # first step is starting index
```

```
print(list2[1:11:3]) # second step is ending index
```

```
# third step is jump
```

Concatenation :

```
List1= [1,2,3,4]
```

```
List2= [5,6,7]
```

```
concatenate=List1+List2
```



Print(concatenate) # output :[1,2,3,4,5,6,7]

2. Tuple :-

Tuple have the same function as the list but the key difference tuples are immutable means unchangeable after creating it. We can also store different data type in a list enclosed in parentheses () and separated by commas.

Example:

```
Friends = ("Ahmad", "Rayan", "Ali")
```

```
Print (Friends) # output : ("Ahmad", "Rayan", "Ali")
```

➤ **Some operations in tuple :**

Here are some operations in tuple

- **Slicing:**

```
num = (1,2,3,4,5,6)
```

```
print (num [2:5]) # output: (3,4,5)
```

```
print (num [-1]) # output: (6)
```

- **finding the elements in tuple:**

```
number=(12,23,4,68,45)
```

```
if 45 in tuple:
```

```
    print("45 is present in number at index", number.index(45))
```

```
else:
```

```
    print("45 is not present in tuple")
```

output:

```
45 is present in number at index , 4
```

- **More operations :-**

Tuple = (2,4,3,8,5,4,6,3,6,634,45,0)



```
print (len(Tuple))      # it shows the length of tuple
print (Tuple.count(6))  # it counts the number that how many time it repeats
print (Tuple.index(634)) # it shows the index value of number
print ("3 at index after 4th index :",Tuple.index(3,4,8))

# -> first point is value to be searched
# -> second point is starting index
# -> third point is ending index
```

3. Sets:-

Sets are used to store multiple items in a single variable. A set is a collection which is *unordered*, *unchangeable**, and *unindexed*. Set *items* are unchangeable, but you can remove items and add new items. It can store all type of data. Sets are close in curly braces {}.

Example:

```
info = {"author", 10, 34.4, True, 10}
print(info)
```

Properties:-

1) Unordered

Unordered means that the items in a set do not have a defined order.

2) Unchangeable

Set items are unchangeable, meaning that we cannot change the items after the set has been created.

Once a set is created, you cannot change its items, but you can remove items and add new items.

3) Duplicates Not Allowed

Sets cannot have two items with the same value.

Example:



```
num = {1,2,3,4,56,2,7,2}
print(num)

# output : {1, 2, 3, 4, 7, 56}
```

➤ SETS METHODS AND OPERATIONS

1. Union and update:

It join the two sets.

Example:

union inn sets join two sets

```
countries1={"Pakistan","India","China"}
```

```
countries2={"USA","UK","China"}
```

```
print(countries1.union(countries2))
```

update method also joins two sets

```
countries1.update(countries2)
```

```
print(countries1)
```

2. Intersection and update:

It also join the two sets with common numbers in it.

Example:

intersection in sets gives common values

```
cities1 = {"Karachi", "Lahore", " Multan "}
```

```
cities2 = {"Karachi", "Islamabad"," Multan "}
```

```
print(cities1.intersection(cities2))
```

intersection_update method also gives common values

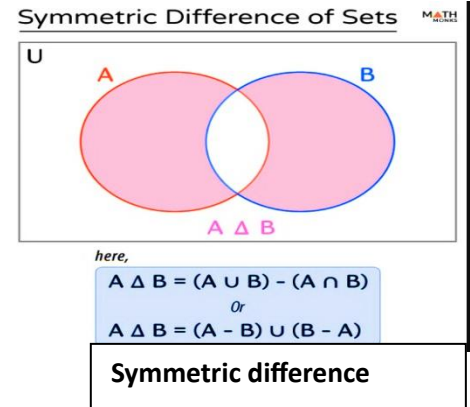
```
cities1.intersection_update(cities2)
```



```
print(cities1)
```

3. Symmetric difference:

Symmetric difference gives values that are not common in both sets.



Example:

```
friends1={"Ahmad","Rayan","Rafy","Faizan"}  
friends2={"Asad","Hussnain","Rayan","Rafy"}  
friends=friends1.symmetric_difference(friends2)  
friends1.symmetric_difference_update(friends2)  
print(friends)  
print(friends1)
```

4. Difference:

difference gives values that are not common in both sets but only from first set

Formula is $A - B$

Example:

```
A={"A","B","C","D"}  
B={"A","E","F","G"}  
c=A.difference(B)  
B.difference_update(A)  
print(c)  
print(B)
```

5. Is super set:

Issuperset method returns true if all values of set B are present in set A.

Example:

```
value1={1,2,3,4,5}
```



```
value2={1,2,3,4}
```

```
print(value1.issuperset(value2))
```

6. Is sub set:

issubset method returns true if all values of set A are present in set B.

Example:

```
num1={1,2,3,4,5,6}
```

```
num2={1,2,3,4,5}
```

```
print(num2.issubset(num1))
```

7. Add item:

You can add item in set through this syntax.

Example:

```
cousins={"Huzaifa","Hanzala","Adil","Moiz"}
```

```
cousins.add("Hamza")
```

```
print(cousins)
```

8. Remove | discard item:

Remove function remove the item in the set and discard also has same function but the key difference is discard doesn't generate error if item is not present in a set.

Example:

```
cousins={"Huzaifa","Hanzala","Adil","Moiz"}
```

```
cousins.remove("Moiz")
```

```
print(cousins)
```

```
cousins.discard("ali")
```

```
print(cousins)
```

```
# pop method
```

```
item=cousins.pop()
```

```
print("The remove item from the set is ",item)
```

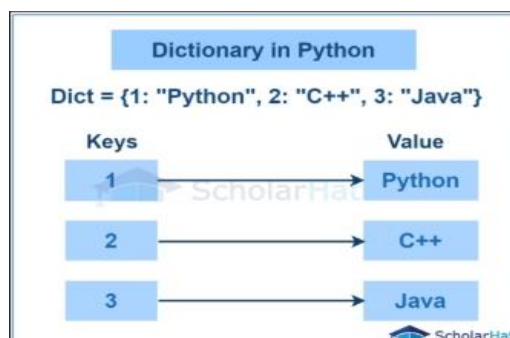
```
print(cousins)
```



4. Dictionaries:

Dictionaries are used to store data value in key:value pair. It is a collection which is *ordered, mutable and does not allow duplicates.

Dictionaries are written with curly brackets, and have keys and values:



Example:

```
dic = {  
    "Name" : "Abdul Rehman",  
    "Age" : 17,  
    "city" : "Chishtian",  
    "ph_num": "0307 4809035"  
}  
print(dic)
```

Properties:

1. It is mutable which means that changeable.
2. It is in ordered.
3. It doesn't allow duplicates.

Operations in dictionaries:

❖ Accessing items:

```
dict = {  
    "name": "Dictionary Method",
```



```
"age": 1,  
"city": "Internet",  
"type": "Data Structure"  
}
```

```
print(dic.items())
```

❖ **Accessing keys:**

```
dict = {  
    "name": "Dictionary Method",  
    "age": 1,  
    "city": "Internet",  
    "type": "Data Structure"  
}
```

```
print(dic.keys())
```

❖ **Accessing values:**

```
print(dic.values())
```

More methods in dictionaries:

■ **Update method:**

update method is used to add the elements of one dictionary to another dictionary

Example:

```
std1={  
    "Mana": 89,  
    "Ahmad": 78,  
    "Faizan": 91,  
    "Rayan": 63  
}  
std2 = {  
    "Ayan": 88,  
    "Dawood": 95,  
    "umair": 79  
}
```

```
std1.update(std2)
```

```
print(std1)
```



▪ Clear method :

clear method is used to remove all the elements from the dictionary

Example:

```
std1={  
    "Mana": 89,  
    "Ahmad": 78,  
    "Faizan": 91,  
    "Rayan": 63  
}  
std2 = {  
    "Ayan":88,  
    "Dawood":95,  
    "umair":79  
}  
std2.clear()  
print(std2)
```

▪ pop and del method:

pop method removes the item with the specified key name . popitem method removes the last inserted item from the dictionary.

Example:

```
std1={  
    "Mana": 89,  
    "Ahmad": 78,  
    "Faizan": 91,  
    "Rayan": 63  
}  
std2 = {  
    "Ayan":88,  
    "Dawood":95,  
    "umair":79  
}  
std1.pop("Mana")
```

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ABDUL REHMAN AZAM

```
print(std1)
```

```
std1.popitem()
```

```
print(std1)
```

del :

del method removes the dictionary completely

above the same example :

```
del std2
```

```
print(std2)
```

it will give error because std2 is deleted

but you can also use del method to remove a specific item from the dictionary

```
del std1["Dawood"]
```

```
print(std1)
```

THE END

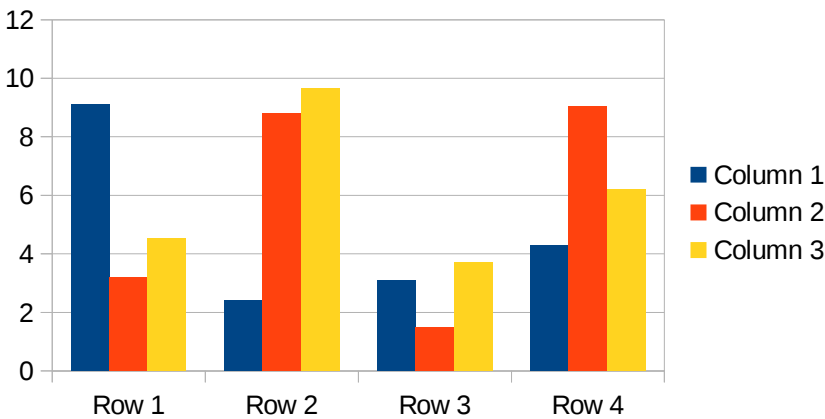
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Sample PDF Document

Robert Maron
Grzegorz Grudziński

February 20, 1999

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Chapter 1

Template

1.1 How to compile a .tex file to a .pdf file

1.1.1 Tools

To process the files you (may) need:

- `pdflatex` (for example from `tetex` package $\geq 0.9-6$, which you can get from [Red Hat 5.2](#));
- `acroread` (a PDF viewer, available from <http://www.adobe.com/>);
- `ghostscript` ≥ 5.10 (for example from [Red Hat Contrib](#)) and `ghostview` or `gv` (from RedHat Linux);
- `efax` package could be useful, if you plan to fax documents.

1.1.2 How to use the tools

Follow these steps:

1. put all source .tex files in one directory, then `chdir` to the directory (or put some of them in the `LATEX` search path — if you know how to do this);
2. run “`pdflatex file.tex`” on the main file of the document three times (three — to prepare valid table of contents);
3. to see or print the result use `acroread` (unfortunately some versions of `acroread` may produce PostScript which is too complex), or

4. run `ghostscript`: “`gv file.pdf`” to display or:
“`gs -dNOPAUSE -sDEVICE=pswrite -q -dBATCH -sOutputFile=file.ps file.pdf`”
to produce a PostScript file;
5. run “`fax send phone-number file.ps`” as root to send a fax, or — if you know how to do this — modify the `fax` script to be able to fax `.pdf` files directly (you have to insert “`|%PDF*`” somewhere...).

1.2 How to write a document

1.2.1 The main document

Choose the name of the document, say `document`. Copy `template.tex` to `document.tex`, then edit it, change the title, the authors and set proper `include(s)` for all the chapters.

1.2.2 Chapters

Each chapter should be included in the main document as a separate file. You can choose any name for the file, but we suggest adding a suffix to the name of the main file. For our example we use the file name `document_chapter1.tex`.

First, copy `template_chapter.tex` to `document_chapter1.tex` and add the line

```
\include{document_chapter1}
```

in the `document.tex`, then edit `document_chapter1.tex`, change the chapter title and edit the body of the chapter appropriately.

1.2.3 Spell-checking

Do use a spell-checker, please!

You may also want to check grammar, style and so on. Actually you should do it (if you have enough spare time). But you *must* check spelling!

You can use the `ispell` package for this, from within `emacs`, or from the command line:

```
ispell -t document_chapter1.tex
```

1.3 $\text{\LaTeX}$ and $\text{pdf}\text{\LaTeX}$ capabilities

1.3.1 Overview

First you edit your source `.tex` file. In $\text{\LaTeX}$ you compile it using the `latex` command to a `.dvi` file (which stands for device-independent). The `.dvi` file can be converted to any device-dependent format you like using an appropriate driver, for example `dvips`.

When producing `.pdf` files you should use `pdflatex`, which produces directly `.pdf` files out of `.tex` sources. Note that in the `.tex` file you may need to use some PDF specific packages.

For viewing `.tex` files use your favourite text editor, for viewing `.dvi` files under X Window System use `xdvi` command, `.ps` files can be viewed with `gv` (or `ghostview`) and `.pdf` files with `acroread`, `gv` or `xpdf`.

1.3.2 $\text{\LaTeX}$

A lot of examples can be found in this document.

You should also print

- `doc/latex/general/latex2e.dvi` and
- `doc/latex/general/lshort2e.dvi`

from your `tetex` distribution (usually in

- `/usr/share/texmf` or
- `/usr/lib/texmf/texmf`).

1.3.3 $\text{pdf}\text{\LaTeX}$

Consult `doc/pdftex/manual.pdf` from your `tetex` distribution for more details. Very useful informations can be found in the `hyperref` and `graphics` package manuals:

- `doc/latex/hyperref/manual.pdf` and
- `doc/latex/graphics/grfguide.dvi`.

1.3.4 Examples

References

MIMUW

Hyperlinks

This is a target.

And [this is a link](#).

Dashes, etc.

There are three kinds of horizontal dash:

- - (use inside words; for example “home-page”, “X-rated”)
- – (use this one between numbers; for example “pages 2–22”)
- — (use this one as a sentence separator — like here)

National characters

- ó, é, í, ...
- è, à, ì, ...
- ô, ê, ...
- ã, ñ, ...
- ö, ë, ...
- ž
- æ, ø
- ł, ø, ß

There are other ways to do this, see the documentation for `inputenc` package.

Reserved characters

Some characters have some special meaning, thus cannot be entered in the usual way.

- \$ & % # _ { }
- \
- ~ ^

Math

- $1^2, 1^{2n}, \dots$
- $i_1, i_{2n}, \dots$
- $\frac{1}{2}, \frac{2n}{2-3}, \dots$
- $\alpha, \beta, \gamma, \Omega, \dots$
- $\rightarrow, \Rightarrow, \geq, \neq, \in, \star, \dots$
- $\sqrt{2}, \dots$
- $\overline{2+2}, \dots$

For more examples and symbols see chapter 3 of `lshort2e.dvi`.

Fonts

- Roman
- *Emphasis*
- Medium weight — the default
- **Boldface**
- Upright
- *Slanted*
- Sans serif
- SMALL CAPS
- Typewriter
- and sizes:
 - tiny
 - scriptsize
 - footnotesize
 - small
 - normalsize

- large
- Large
- LARGE
- huge
- Huge

Sample PDF

This is a simple PDF file. Fun fun fun.

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